

Beyond Olive's Hybrid Math

By AlienJon

Ok so I've been working with a more in-depth model for "hybrids" "composite spinning" whatever you want to call it... It's all just relations in configuration space. At any rate its starting to gel to where I can talk about it semi-cohesively, thanks to playing with Cat Eyes as of late. So if you aren't already familiar with Olive's post about possible hybrid combinations here's your homework:

...doing the math on hybrids:

techpoi.tribe.net/thread/78...d54c0dc644

Weeeellll, his math was incomplete (or maybe I just like to over complicate things). So, I'm really starting to get comfortable with spinning a Cat Eye. Cat eye is Arash'ese for a 1beat antispin ellipse:

techpoi.tribe.net/thread/55...60ee9f6b34

Zan post as well:

techpoi.tribe.net/thread/6d...6bac1a188

As a result, I've been thinking of all those Olive hybrids in terms of one beat and unit circle. The poi unit circle idea is that your poi length = 2units, in other words, poi length is the diameter of the circle an isolation makes. Same thing for the extension that can be substituted in for an iso hybrid style... and same thing for your hands path in a cat eye. I can't yet do a horizontal cat eye, but I think it's possible.

Anyhow, once I started analyzing all those Olive hybrids as unit circle one-beats instead of polyrhythms, The relational poi thing started to become way more clear to me. *deep breath*

So you have to look at the relation of:

- a) Lhand vs Lpoi [0,0]
- b) Rhand vs Rpoi [0,0]
- c) Lpoi vs Rpoi [0,0]
- d) Lhand vs Rhand [0,0]
- e) Lhand vs Rpoi [0,0]
- f) Rhand vs Lpoi [0,0]

Each relation has 2 binary variables:

Orientation (spin)= same[0] or opposite[1]

Phase (timing)= same[0] or opposite[1] (I'm leaving quarter-time and polyrhythmn out for now, more complex model later)

So as far as hand orientation and phase, it's determined by a vector pointing through the hand, away from the pivot for the arm; just like for the poi it is determined by a vector through the head, pointing away from the pivot as it spins. When the vector has cycled through all orientations in the spin plane, that is one beat... or cycle.

Now if you have the same values in a-f, you have a high level of symmetry in poi "configuration space". This gives you 4 different non-hybrid movements:

- same-time extensions
- same-time isolations
- split-time extensions
- split-time isolations

The 4 "same poi vs same hand" relationships determine the unit circle "driving styles":

- [0,0]=extension
- [0,1]=isolation
- [1,0]=vertical cat eye
- [1,1]=horizontal cat eye

We are all familiar with the 4 poi vs poi and hand vs hand spin but lets write it out anyway:

- same direction, same time [0,0]
- same direction, split time [0,1]
- opposite direction, same time [1,0]
- opposite direction, split time [1,1]

In Olives model he applied the 4 to hand/arm motion, but only direction and not timing to poi motion, assuming that they naturally fit in certain ways. In the case of some of the hybrids with more uniform configurations, if you change the poi timing they turn into one of the 4 non-hybrids from above. However, the more complex ones have different configurations based on timing: ie a vertical cat eye as opposed to a horizontal cat eye.

I'm still digesting and grokking the importance of poi vs opposite hand, as something more than an epiphenomena, but it feels like it is another angle that is pertinent.

Ultimately this the beginning of a model to composite movements together that encompasses and goes beyond hybrids. I'd like to try and make it extensible. It should include stalls, compound timings and hence compound circles, linear movements as well as the circular movement, polyrhythms, etc. This should lay the foundation to describe other driving styles such as spin/antispin, pundula, whips, parallels, etc.

I'm working on animating all this stuff. I have the original Olive list transposed to unit circle 1-beats. I need to render it and compile a video... and of course start to expand it exhaustively to show all of the combinations, with an attribute overlay, so people can start to see the relations at a glance.

Ok 4:20am,
-Alien signing off